Antibiotic Bias Research

Nathan Seaney

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Overview

This study investigates whether differences in antibiotic control across microbiome studies may introduce systematic bias in observed microbial associations with colorectal cancer (CRC).

Specifically, we compare:

- Studies that explicitly controlled for antibiotic usage
- Studies that did not report or control for antibiotic usage

The primary objective is to evaluate whether microbial signals associated with early versus late onset CRC remain consistent across these study groups.

---

2. Data Loading

Publicly available metagenomic datasets were obtained using the `curatedMetagenomicData` package. These datasets contain species-level relative abundance profiles along with associated clinical metadata.

All datasets were loaded and stored in a unified list structure for downstream processing. —

2. Data Loading Chunk

```
# Define studies
studies <- c(
  "ZellerG_2014",
  "FengQ_2015",
  "YuJ_2015",
  "WirbelJ_2018",
  "ThomasAM_2019_c",
  "VogtmannE_2016",
  "YachidaS_2019",
  "GuptaA_2019",
  "HanniganGD_2017"
)
```

```r
# cat("\n=== Loading datasets ===\n")

data_list <- lapply(studies, function(study) {
  curatedMetagenomicData(
    paste0(study, ".relative_abundance"),
    dryrun = FALSE
  )[[1]]
})
```

3. Preprocessing

To ensure comparability across studies, species abundance matrices were standardized by aligning all datasets to a common set of microbial features.

Metadata tables were harmonized by:

- Adding missing variables as `NA`
- Ensuring consistent column ordering
- Tracking study origin for each sample

Samples were restricted to those labeled as colorectal cancer (CRC), and patients were stratified into two groups:

- Early onset CRC (age > 50)
- Late onset CRC (age ≥ 50)

---

3. Preprocessing Chunk

Everything from: - species extraction
- metadata extraction
- alignment
- CRC filtering

goes here.

```r
# (your sections 6-10 go here exactly as written)

### ---- 6. Extract species + metadata ----
species_list <- lapply(data_list, function(tse) {
  mat <- t(assay(tse, "relative_abundance"))
  colnames(mat) <- sub(".*s__", "", colnames(mat))  # clean names
  mat
})

### ---- 6b. Extract metadata (REQUIRED) ----
```

```r
meta_list <- lapply(seq_along(data_list), function(i) {
  m <- as.data.frame(colData(data_list[[i]]))
  m$study <- studies[i]   # track dataset origin
  m
})

### ---- 7. Align species across datasets (FIXED) ----

cat("\n=== Aligning species ===\n")
```

```
##
## === Aligning species ===
```

```r
all_species <- unique(unlist(lapply(species_list, colnames)))

species_list <- lapply(species_list, function(mat) {

  # Ensure matrix
  mat <- as.matrix(mat)

  # Add missing columns safely
  missing <- setdiff(all_species, colnames(mat))

  if (length(missing) > 0) {
    add_mat <- matrix(0,
                      nrow = nrow(mat),
                      ncol = length(missing),
                      dimnames = list(rownames(mat), missing))
    mat <- cbind(mat, add_mat)
  }

  # Reorder columns to match reference
  mat[, all_species, drop = FALSE]
})

### ---- 8. Combine into one dataset (FIXED) ----

# Find all metadata columns across datasets
all_meta_cols <- unique(unlist(lapply(meta_list, colnames)))

# Add missing columns to each dataset
meta_list <- lapply(meta_list, function(df) {
  missing <- setdiff(all_meta_cols, colnames(df))
  for (col in missing) {
    df[[col]] <- NA
  }
  df[, all_meta_cols]  # enforce same column order
})

# Now safely combine
species_all <- do.call(rbind, species_list)
meta_all    <- do.call(rbind, meta_list)
```

```r
dim(meta_all)
```

```
## [1] 1510   43
```

```r
invisible(table(meta_all$study))
table(meta_all$disease)
```

```
##
##                                adenoma                   adenoma;fatty_liver
##                                    143                                    12
##     adenoma;fatty_liver;hypertension                  adenoma;hypertension
##                                     14                                     8
##            carcinoma_surgery_history                                    CRC
##                                     40                                   578
##                       CRC;fatty_liver         CRC;fatty_liver;hypertension
##                                      3                                    12
##                      CRC;hypertension                               CRC;T2D
##                                     12                                    29
##     CRC;T2D;fatty_liver;hypertension                  CRC;T2D;hypertension
##                                      4                                     2
##                           fatty_liver             fatty_liver;hypertension
##                                      8                                    13
##                            few_polyps                               healthy
##                                      5                                   582
##                          hypertension                                   T2D
##                                     11                                    16
##                           T2D;adenoma              T2D;adenoma;fatty_liver
##                                      1                                     1
## T2D;adenoma;fatty_liver;hypertension             T2D;adenoma;hypertension
##                                      1                                     2
##                       T2D;fatty_liver         T2D;fatty_liver;hypertension
##                                      3                                     9
##                      T2D;hypertension
##                                      1
```

```r
### ---- 8b. Clean disease labels (REQUIRED FIX) ----

meta_all$disease_clean <- sub(";.*", "", meta_all$disease)

cat("\n=== Cleaned disease labels ===\n")
```

```
##
## === Cleaned disease labels ===
```

```r
print(table(meta_all$disease_clean))
```

```
##
##                    adenoma carcinoma_surgery_history                       CRC
##                        177                         40                       640
##                fatty_liver                 few_polyps                   healthy
```

```
##                        21                        5                        582
##              hypertension                      T2D
##                        11                       34
```

```
### ---- 9. Keep only CRC samples (FIXED) ----

crc_idx <- grepl("^CRC", meta_all$disease)

species_crc <- species_all[crc_idx, ]
meta_crc    <- meta_all[crc_idx, ]

cat("\n=== CRC samples ===\n")
```

```
##
## === CRC samples ===
```

```
print(nrow(meta_crc))
```

```
## [1] 640
```

```
### ---- 10. Define early vs late CRC ----
meta_crc$age_group2 <- ifelse(meta_crc$age < 50, "early", "late")

cat("\n=== Age groups ===\n")
```

```
##
## === Age groups ===
```

```
print(table(meta_crc$age_group2))
```

```
##
## early  late
##    78   562
```

4. Core Analysis

We first evaluated microbial dysbiosis using the presence and abundance of *Fusobacterium*, a genus strongly associated with CRC.

Two complementary analyses were performed:

- Presence/absence comparison using Fisher's exact test
- Abundance comparison using the Wilcoxon rank-sum test

To identify differentially abundant species between early and late CRC, we applied a linear modeling framework using the `limma` package. Models were adjusted for study-specific effects to account for batch variability.

Statistical significance was determined using false discovery rate (FDR) correction. —

4. Core Analysis (Fusobacterium + limma)

```r
### ---- 11. Define dysbiosis (Fusobacterium presence) ----

# Use ALL Fusobacterium species (stronger than single species)
fus_cols <- grep("Fusobacterium", colnames(species_crc))

if (length(fus_cols) == 0) {
  stop("No Fusobacterium species found in dataset")
}

# Dysbiosis = presence of ANY Fusobacterium
dysbiosis <- rowSums(species_crc[, fus_cols, drop = FALSE]) > 0


### ---- 12. Compare early vs late CRC (presence) ----

cat("\n=== Dysbiosis (presence) ===\n")
```

```
##
## === Dysbiosis (presence) ===
```

```r
print(table(dysbiosis, meta_crc$age_group2))
```

```
##
## dysbiosis early late
##     FALSE    45  260
##     TRUE     33  302
```

```r
cat("\n=== Proportions ===\n")
```

```
##
## === Proportions ===
```

```r
print(prop.table(table(dysbiosis, meta_crc$age_group2), 2))
```

```
##
## dysbiosis     early      late
##     FALSE 0.5769231 0.4626335
##     TRUE  0.4230769 0.5373665
```

```r
cat("\n=== Fisher test ===\n")
```

```
##
## === Fisher test ===
```

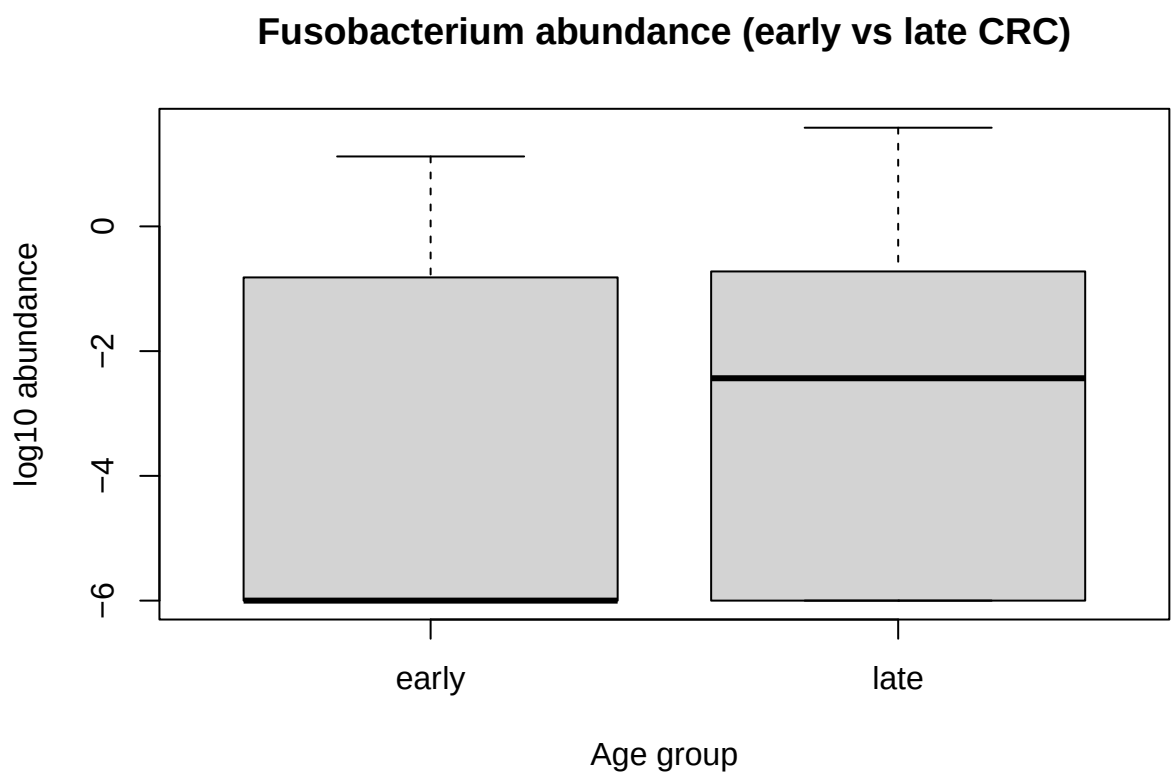
```r
print(fisher.test(table(dysbiosis, meta_crc$age_group2)))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(dysbiosis, meta_crc$age_group2)
## p-value = 0.06932
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.9559522 2.6433569
## sample estimates:
## odds ratio
##   1.582753
```

```r
### ---- 13. Compare abundance (stronger analysis) ----

# Total Fusobacterium abundance per sample
fus_abundance <- rowSums(species_crc[, fus_cols, drop = FALSE])

boxplot(log10(fus_abundance + 1e-6) ~ meta_crc$age_group2,
        main = "Fusobacterium abundance (early vs late CRC)",
        ylab = "log10 abundance",
        xlab = "Age group")
```




```r
# Optional: Wilcoxon test (better for abundance)
cat("\n=== Wilcoxon test (abundance) ===\n")
```

```
##
```

```
## === Wilcoxon test (abundance) ===
```

```r
print(wilcox.test(fus_abundance ~ meta_crc$age_group2))
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  fus_abundance by meta_crc$age_group2
## W = 19684, p-value = 0.1222
## alternative hypothesis: true location shift is not equal to 0
```

```r
### ---- 14. Clean filtered metadata (FIXED) ----

meta <- meta_all %>%
  filter(
    study %in% studies,
    grepl("^CRC|healthy", disease),  # keep CRC + controls
    !is.na(age)
  )

if ("antibiotics_current_use" %in% colnames(meta)) {
  cat("=== antibiotics_current_use ===\n")
  print(table(meta$study, meta$antibiotics_current_use, useNA = "always"))
} else {
  cat("No antibiotics_current_use column found\n")
}
```

```
## === antibiotics_current_use ===
##
##                    no <NA>
##   FengQ_2015       62    0
##   GuptaA_2019      60    0
##   HanniganGD_2017  55    0
##   ThomasAM_2019_c   0   80
##   VogtmannE_2016    0  104
##   WirbelJ_2018      0  125
##   YachidaS_2019     0  504
##   YuJ_2015        112    0
##   ZellerG_2014      0  114
##   <NA>              0    0
```

```r
if ("antibiotics_family" %in% colnames(meta)) {
  cat("\n=== antibiotics_family ===\n")
  print(table(meta$study, meta$antibiotics_family, useNA = "always"))
} else {
  cat("No antibiotics_family column found\n")
}
```

```
## No antibiotics_family column found
```

```r
### ---- 15. Differential analysis: early vs late CRC (FIXED + CLR) ----

cat("\n=== Running differential analysis (limma with CLR) ===\n")
```

```
##
## === Running differential analysis (limma with CLR) ===
```

```r
# ---- 15.1 Remove zero-variance species ----
keep <- apply(species_crc, 2, var, na.rm = TRUE) > 0
species_crc_filt <- species_crc[, keep]

cat("Kept", ncol(species_crc_filt), "species after filtering\n")
```

```
## Kept 811 species after filtering
```

```r
# ---- 15.2 CLR TRANSFORMATION (ADDED FIX) ----

# Add pseudocount
species_crc_clr <- species_crc_filt + 1e-6

# Apply CLR row-wise
species_crc_clr <- t(apply(species_crc_clr, 1, function(x) {
  log(x) - mean(log(x))
}))


# ---- 15.3 Ensure factor levels ----
meta_crc$age_group2 <- factor(meta_crc$age_group2,
                              levels = c("late", "early"))
meta_crc$study <- factor(meta_crc$study)


# ---- 15.4 Design matrix ----
design <- model.matrix(~ age_group2 + study, data = meta_crc)


# ---- 15.5 Run limma ON CLR DATA ----
library(limma)

fit <- lmFit(t(species_crc_clr), design)
fit <- eBayes(fit)


# ---- 15.6 Extract results ----
res_limma <- topTable(
  fit,
  coef = "age_group2early",
  number = Inf,
  sort.by = "P"
)
```

```r
# ---- 15.7 Add interpretation ----
res_limma$species <- rownames(res_limma)

res_limma$direction <- ifelse(
  res_limma$logFC > 0,
  "higher_in_early",
  "higher_in_late"
)


# ---- 15.8 Output ----
cat("\n=== Top differentially abundant species (CLR) ===\n")
```

```
##
## === Top differentially abundant species (CLR) ===
```

```r
print(head(res_limma[, c("species", "logFC", "P.Value", "adj.P.Val", "direction")], 20))
```

```
##                                                    species      logFC
## Dialister_micraerophilus             Dialister_micraerophilus  0.6178034
## Bifidobacterium_dentium               Bifidobacterium_dentium -2.2809907
## Escherichia_coli                             Escherichia_coli -2.7205191
## Gemella_asaccharolytica               Gemella_asaccharolytica  0.5761104
## Anaerotruncus_colihominis           Anaerotruncus_colihominis  2.0665094
## Pseudoflavonifractor_sp_An184   Pseudoflavonifractor_sp_An184  1.4405494
## Streptococcus_sp_HMSC070B10       Streptococcus_sp_HMSC070B10  0.2197690
## Haemophilus_parainfluenzae         Haemophilus_parainfluenzae -2.0311340
## Peptoniphilus_harei                       Peptoniphilus_harei  0.6628793
## Mogibacterium_diversum                 Mogibacterium_diversum -1.6570184
## Peptostreptococcus_anaerobius   Peptostreptococcus_anaerobius  1.1471256
## Anaeromassilibacillus_sp_An250 Anaeromassilibacillus_sp_An250  1.6755652
## Selenomonas_sp_oral_taxon_920   Selenomonas_sp_oral_taxon_920  0.2220608
## Megasphaera_micronuciformis       Megasphaera_micronuciformis -1.2765640
## Parvimonas_sp_oral_taxon_110     Parvimonas_sp_oral_taxon_110  0.3811615
## Oscillibacter_sp_57_20                 Oscillibacter_sp_57_20  2.0919414
## Gardnerella_vaginalis                   Gardnerella_vaginalis  0.3162329
## Aeromonas_media                               Aeromonas_media  0.2980126
## Actinomyces_sp_HPA0247                 Actinomyces_sp_HPA0247 -1.0249345
## Bilophila_wadsworthia                   Bilophila_wadsworthia  1.6166872
##                                   P.Value   adj.P.Val       direction
## Dialister_micraerophilus       3.966326e-06 0.003216691 higher_in_early
## Bifidobacterium_dentium        3.535588e-05 0.013023242  higher_in_late
## Escherichia_coli               4.817475e-05 0.013023242  higher_in_late
## Gemella_asaccharolytica        1.988801e-04 0.040322938 higher_in_early
## Anaerotruncus_colihominis      3.447324e-04 0.055915595 higher_in_early
## Pseudoflavonifractor_sp_An184  5.344768e-04 0.072243447 higher_in_early
## Streptococcus_sp_HMSC070B10    8.636522e-04 0.100060278 higher_in_early
## Haemophilus_parainfluenzae     1.040565e-03 0.105487315  higher_in_late
## Peptoniphilus_harei            1.320251e-03 0.118969276 higher_in_early
## Mogibacterium_diversum         1.536967e-03 0.124648015  higher_in_late
## Peptostreptococcus_anaerobius  2.108655e-03 0.152199580 higher_in_early
## Anaeromassilibacillus_sp_An250 2.252028e-03 0.152199580 higher_in_early
```

```
## Selenomonas_sp_oral_taxon_920  3.127477e-03 0.175415438 higher_in_early
## Megasphaera_micronuciformis    3.679186e-03 0.175415438  higher_in_late
## Parvimonas_sp_oral_taxon_110   3.837229e-03 0.175415438 higher_in_early
## Oscillibacter_sp_57_20         4.048615e-03 0.175415438 higher_in_early
## Gardnerella_vaginalis          4.065980e-03 0.175415438 higher_in_early
## Aeromonas_media                4.100334e-03 0.175415438 higher_in_early
## Actinomyces_sp_HPA0247         4.114942e-03 0.175415438  higher_in_late
## Bilophila_wadsworthia          4.325905e-03 0.175415438 higher_in_early
```

```r
cat("\n=== Significant species (FDR < 0.05) ===\n")
```

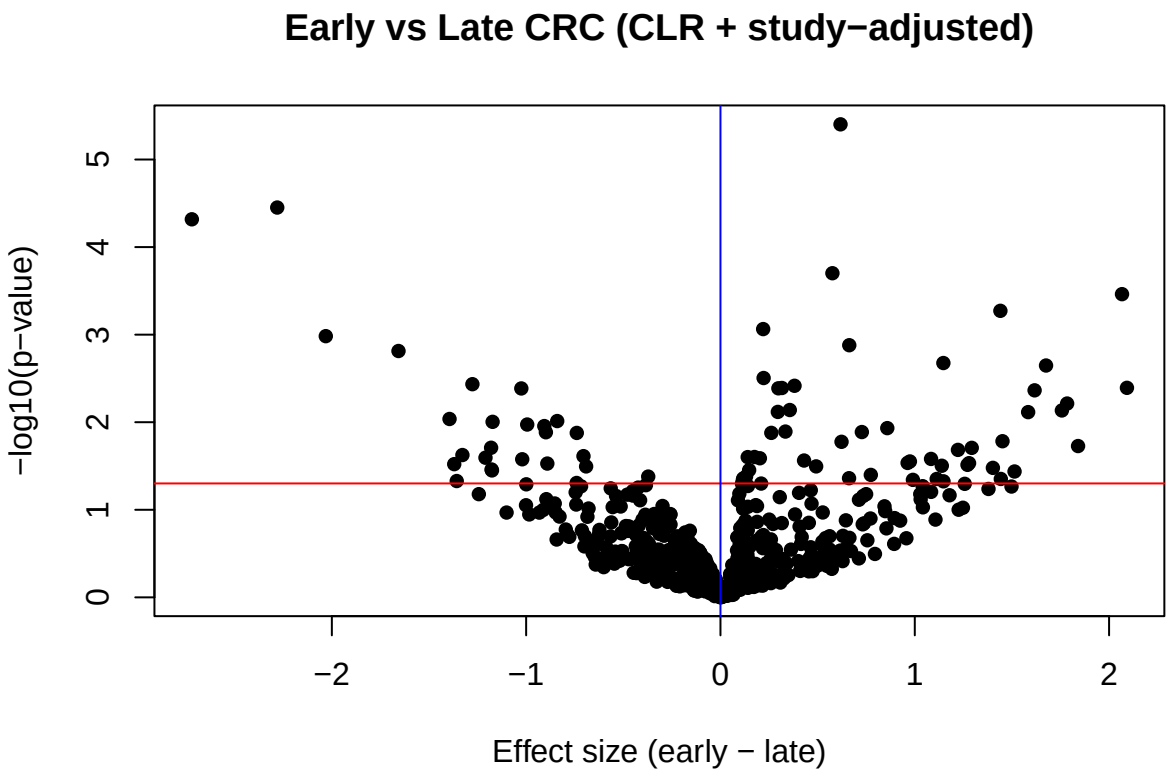
```
##
## === Significant species (FDR < 0.05) ===
```

```r
print(sum(res_limma$adj.P.Val < 0.05, na.rm = TRUE))
```

```
## [1] 4
```

```r
# ---- 15.9 Volcano plot ----
plot(
  res_limma$logFC,
  -log10(res_limma$P.Value),
  pch = 16,
  main = "Early vs Late CRC (CLR + study-adjusted)",
  xlab = "Effect size (early - late)",
  ylab = "-log10(p-value)"
)

abline(h = -log10(0.05), col = "red")
abline(v = 0, col = "blue")
```

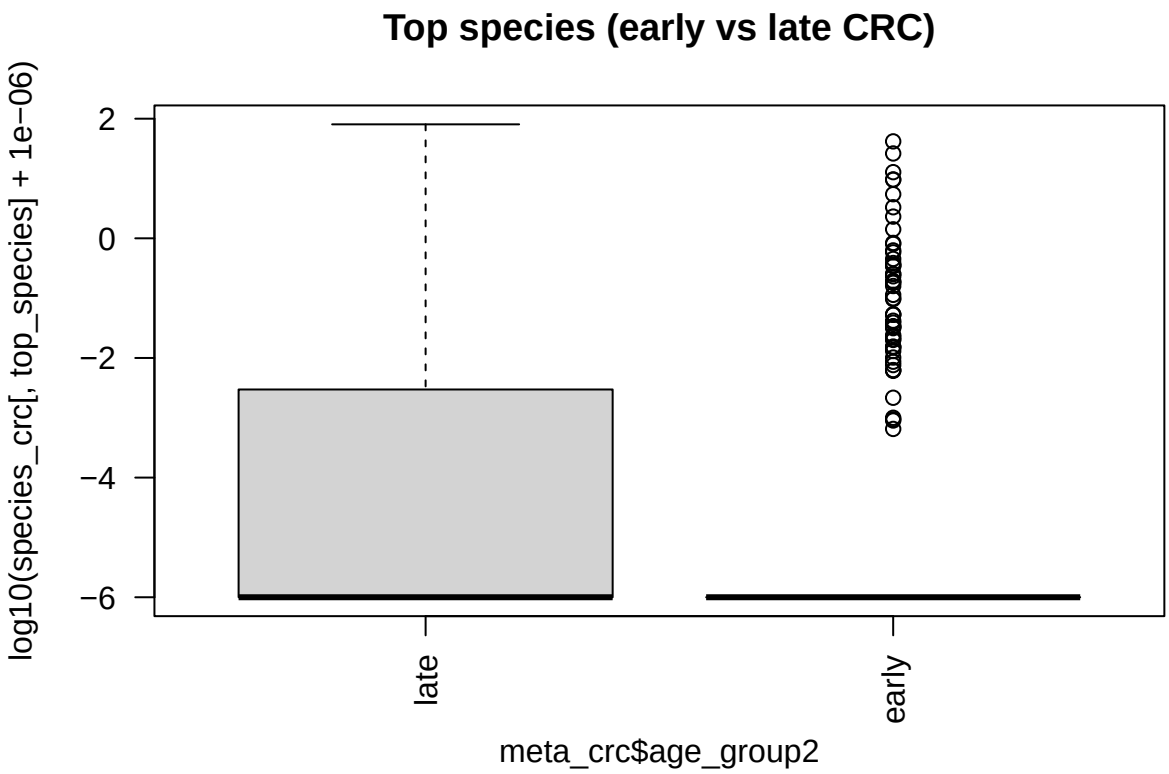



```r
# ---- 15.10 Downstream summaries ----
sig <- res_limma %>% filter(adj.P.Val < 0.05)

early_enriched <- sig %>% filter(logFC > 0)
late_enriched  <- sig %>% filter(logFC < 0)

top_species <- head(sig$species, 10)

boxplot(
  log10(species_crc[, top_species] + 1e-6) ~ meta_crc$age_group2,
  las = 2,
  main = "Top species (early vs late CRC)"
)
```




```r
# ---- 15.11 Random forest (unchanged) ----
library(randomForest)

rf <- randomForest(
  x = species_crc,
  y = meta_crc$age_group2,
  importance = TRUE
)
```

5. Antibiotic Bias Analysis (Exploritory)

To explore potential differences in microbial associations across study designs, datasets were stratified based on whether antibiotic usage was explicitly reported.

Differential abundance analysis (early vs. late CRC) was performed separately within each group using centered log-ratio (CLR) transformed data.

Effect sizes were compared across groups to assess concordance in observed microbial signals.

This exploratory approach provides an intuitive comparison but does not formally test whether antibiotic status modifies the relationship between microbial composition and disease status. —

5. Antibiotic Bias Analysis (KEY RESULT)

```r
cat("\n=== Antibiotic bias analysis (UPDATED with CLR) ===\n")
```

```
##
## === Antibiotic bias analysis (UPDATED with CLR) ===
```

```r
library(dplyr)
library(limma)

# ---- 16.1 Define grouping (DATA-DRIVEN, FIXED) ----
meta_crc$abx_control <- ifelse(
  !is.na(meta_crc$antibiotics_current_use),
  "reported_no_use",
  "unknown"
)

meta_crc$abx_control <- factor(meta_crc$abx_control)

cat("\nSamples per antibiotic group:\n")
```

```
##
## Samples per antibiotic group:
```

```r
print(table(meta_crc$abx_control))
```

```
##
## reported_no_use         unknown
##             177             463
```

```r
# ---- 16.2 Split data (FIXED LABELS) ----
meta_ctrl <- meta_crc %>% filter(abx_control == "reported_no_use")
meta_unk  <- meta_crc %>% filter(abx_control == "unknown")

# Reset factor levels AFTER subsetting
meta_ctrl$age_group2 <- factor(meta_ctrl$age_group2, levels = c("late", "early"))
meta_unk$age_group2  <- factor(meta_unk$age_group2,  levels = c("late", "early"))

species_ctrl <- species_crc[rownames(meta_ctrl), ]
species_unk  <- species_crc[rownames(meta_unk), ]


# ---- 16.3 Remove zero-variance (robust) ----
species_ctrl <- species_ctrl[, apply(species_ctrl, 2, var, na.rm = TRUE) > 0]
species_unk  <- species_unk[, apply(species_unk, 2, var, na.rm = TRUE) > 0]


# ---- 16.4 CLR TRANSFORMATION (CRITICAL FIX) ----

# Add pseudocount to avoid log(0)
```

```r
species_ctrl_clr <- species_ctrl + 1e-6
species_unk_clr  <- species_unk  + 1e-6

# Apply CLR (row-wise)
species_ctrl_clr <- t(apply(species_ctrl_clr, 1, function(x) {
  log(x) - mean(log(x))
}))

species_unk_clr <- t(apply(species_unk_clr, 1, function(x) {
  log(x) - mean(log(x))
}))


# ---- 16.5 Run limma separately (ON CLR DATA) ----
design_ctrl <- model.matrix(~ age_group2, data = meta_ctrl)
design_unk  <- model.matrix(~ age_group2, data = meta_unk)

cat("\nDesign CTRL:\n")
```

```
##
## Design CTRL:
```

```r
print(colnames(design_ctrl))
```

```
## [1] "(Intercept)"     "age_group2early"
```

```r
cat("\nDesign UNK:\n")
```

```
##
## Design UNK:
```

```r
print(colnames(design_unk))
```

```
## [1] "(Intercept)"     "age_group2early"
```

```r
fit_ctrl <- eBayes(lmFit(t(species_ctrl_clr), design_ctrl))
fit_unk  <- eBayes(lmFit(t(species_unk_clr),  design_unk))


# ---- 16.6 Extract results ----
res_ctrl <- topTable(fit_ctrl, coef = "age_group2early", number = Inf)
res_unk  <- topTable(fit_unk,  coef = "age_group2early", number = Inf)

# Add species column
res_ctrl$species <- rownames(res_ctrl)
res_unk$species  <- rownames(res_unk)


# ---- 16.7 Merge results ----
merged <- merge(
```

```r
  res_ctrl[, c("species", "logFC", "adj.P.Val")],
  res_unk[,  c("species", "logFC", "adj.P.Val")],
  by = "species",
  suffixes = c("_reported", "_unknown")
)


# ---- 16.8 Correlation (key test) ----
cor_val <- cor(
  merged$logFC_reported,
  merged$logFC_unknown,
  use = "complete.obs"
)

cat("\nCorrelation of effect sizes:\n")
```

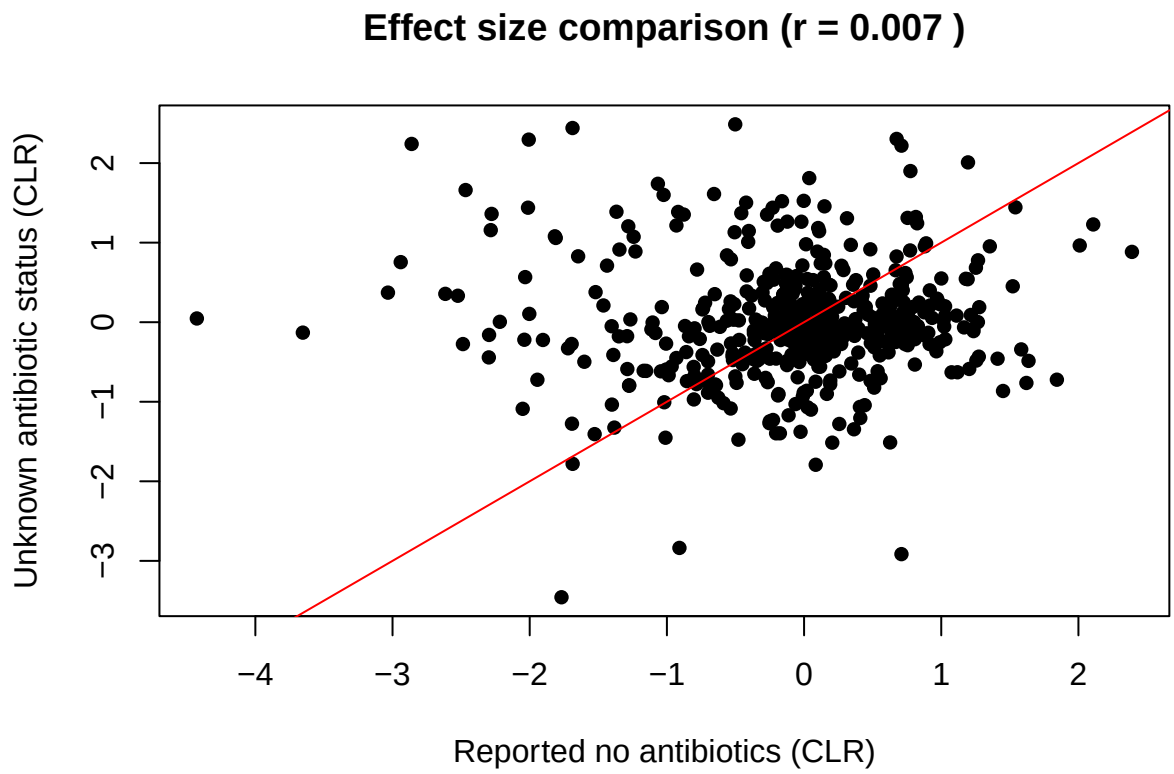
```
##
## Correlation of effect sizes:
```

```r
print(cor_val)
```

```
## [1] 0.006641864
```

```r
# ---- 16.9 Scatter plot ----
plot(
  merged$logFC_reported,
  merged$logFC_unknown,
  pch = 16,
  main = paste("Effect size comparison (r =", round(cor_val, 3), ")"),
  xlab = "Reported no antibiotics (CLR)",
  ylab = "Unknown antibiotic status (CLR)"
)

abline(0, 1, col = "red")
```




```r
# ---- 16.10 Discordant species ----
merged$discordant <- sign(merged$logFC_reported) != sign(merged$logFC_unknown)

discordant_species <- merged %>%
  filter(discordant) %>%
  arrange(desc(abs(logFC_reported - logFC_unknown)))

cat("\nTop discordant species:\n")
```

```
##
## Top discordant species:
```

```r
print(head(discordant_species, 10))
```

```
##                              species logFC_reported adj.P.Val_reported
## 1                Alistipes_putredinis     -2.8608367          0.5303467
## 2              Eubacterium_sp_CAG_180     -4.4265632          0.1451190
## 3  Bifidobacterium_pseudocatenulatum     -2.0071350          0.9679476
## 4            Odoribacter_splanchnicus     -1.6878036          0.9679476
## 5          Parabacteroides_distasonis     -2.4674908          0.5289082
## 6           Bacteroides_xylanisolvens     -2.9403628          0.3317919
## 7                    Prevotella_copri     -2.2778651          0.9207251
## 8          Haemophilus_parainfluenzae      0.7099619          0.9679476
## 9               Bacteroides_stercoris     -2.0123086          0.9679476
## 10           Oscillibacter_sp_CAG_241     -2.2840989          0.8043134
```

```
##    logFC_unknown adj.P.Val_unknown discordant
## 1     2.24071923        0.27417476       TRUE
## 2     0.04661135        0.99962752       TRUE
## 3     2.29505690        0.23980358       TRUE
## 4     2.44081483        0.08954596       TRUE
## 5     1.66100654        0.22638388       TRUE
## 6     0.75494790        0.89476076       TRUE
## 7     1.36098149        0.73795018       TRUE
## 8    -2.91540622        0.01896132       TRUE
## 9     1.43924257        0.70080913       TRUE
## 10    1.15722070        0.71002899       TRUE
```

```r
# ---- 16.11 Compare significance overlap ----
sig_ctrl <- merged %>% filter(adj.P.Val_reported < 0.05)
sig_unk  <- merged %>% filter(adj.P.Val_unknown < 0.05)

cat("\nSignificant (reported):", nrow(sig_ctrl), "\n")
```

```
##
## Significant (reported): 8
```

```r
cat("Significant (unknown):", nrow(sig_unk), "\n")
```

```
## Significant (unknown): 4
```

```r
overlap <- intersect(sig_ctrl$species, sig_unk$species)

cat("Overlap:", length(overlap), "\n")
```

```
## Overlap: 0
```

Interaction-Based Antibiotic Bias Test (Primary Analysis)

To formally evaluate whether antibiotic reporting status modifies microbial associations with early versus late onset colorectal cancer, we fit a single linear model including an interaction term between age group and antibiotic reporting status.

This interaction term directly tests whether the effect of age group (early vs. late CRC) differs depending on whether antibiotic usage was reported.

All analyses were conducted using centered log-ratio (CLR) transformed abundance data and adjusted for study-specific effects.

```r
cat("\n=== Interaction model: age_group × antibiotic status ===\n")
```

```
##
## === Interaction model: age_group × antibiotic status ===
```

```r
library(dplyr)
library(limma)

# ---- 17.1 Define antibiotic grouping ----
meta_crc$abx_control <- ifelse(
  !is.na(meta_crc$antibiotics_current_use),
  "reported_no_use",
  "unknown"
)

meta_crc$abx_control <- factor(meta_crc$abx_control)
meta_crc$age_group2  <- factor(meta_crc$age_group2, levels = c("late", "early"))
meta_crc$study       <- factor(meta_crc$study)

cat("\nSamples per group:\n")
```

```
##
## Samples per group:
```

```r
print(table(meta_crc$age_group2, meta_crc$abx_control))
```

```
##
##         reported_no_use unknown
##   late              159     403
##   early              18      60
```

```r
# ---- 17.2 Filter species ----
keep <- apply(species_crc, 2, var, na.rm = TRUE) > 0
species_crc_filt <- species_crc[, keep]

cat("Kept", ncol(species_crc_filt), "species after filtering\n")
```

```
## Kept 811 species after filtering
```

```r
# ---- 17.3 CLR transformation ----
species_crc_clr <- species_crc_filt + 1e-6

species_crc_clr <- t(apply(species_crc_clr, 1, function(x) {
  log(x) - mean(log(x))
}))


# ---- 17.4 Build interaction model ----
design <- model.matrix(~ age_group2 * abx_control + study, data = meta_crc)

cat("\nDesign matrix:\n")
```

```
##
## Design matrix:
```

```r
print(colnames(design))
```

```
##  [1] "(Intercept)"                        "age_group2early"
##  [3] "abx_controlunknown"                 "studyGuptaA_2019"
##  [5] "studyHanniganGD_2017"               "studyThomasAM_2019_c"
##  [7] "studyVogtmannE_2016"                "studyWirbelJ_2018"
##  [9] "studyYachidaS_2019"                 "studyYuJ_2015"
## [11] "studyZellerG_2014"                  "age_group2early:abx_controlunknown"
```

```r
# ---- 17.5 Run limma ----
fit <- lmFit(t(species_crc_clr), design)
```

```
## Coefficients not estimable: studyZellerG_2014
```

```r
fit <- eBayes(fit)


# ---- 17.6 Extract interaction term ----
res_interaction <- topTable(
  fit,
  coef = "age_group2early:abx_controlunknown",
  number = Inf,
  sort.by = "P"
)

res_interaction$species <- rownames(res_interaction)


# ---- 17.7 Results ----
cat("\n=== Top interaction effects ===\n")
```

```
##
## === Top interaction effects ===
```

```r
print(head(res_interaction[, c("species", "logFC", "P.Value", "adj.P.Val")], 20))
```

```
##                                                    species      logFC
## Streptococcus_sp_HMSC070B10       Streptococcus_sp_HMSC070B10 -0.9198857
## Desulfovibrio_legallii                 Desulfovibrio_legallii -0.7280192
## Acinetobacter_johnsonii               Acinetobacter_johnsonii -0.5783909
## Bifidobacterium_gallinarum         Bifidobacterium_gallinarum -0.9960727
## Bifidobacterium_saeculare           Bifidobacterium_saeculare -1.0909481
## Providencia_rettgeri                     Providencia_rettgeri -0.5412603
## Brevibacterium_aurantiacum         Brevibacterium_aurantiacum -0.4797173
## Brevibacterium_linens                   Brevibacterium_linens -0.4711483
## Escherichia_fergusonii                 Escherichia_fergusonii -0.5052214
## Staphylococcus_warneri                 Staphylococcus_warneri -0.4519535
## Fusobacterium_periodonticum       Fusobacterium_periodonticum -2.4661673
## Haemophilus_parainfluenzae         Haemophilus_parainfluenzae -4.7364163
## Bifidobacterium_pullorum             Bifidobacterium_pullorum -1.1244338
## Leptotrichia_sp_oral_taxon_212 Leptotrichia_sp_oral_taxon_212 -0.5945758
```

```
## Clostridium_sp_CAG_632                 Clostridium_sp_CAG_632  1.2688301
## Citrobacter_freundii                     Citrobacter_freundii -2.3844040
## Lactobacillus_johnsonii                Lactobacillus_johnsonii -0.8938373
## Kosakonia_pseudosacchari              Kosakonia_pseudosacchari -0.4763580
## Citrobacter_youngae                       Citrobacter_youngae -2.1924577
## Streptococcus_massiliensis          Streptococcus_massiliensis -0.5238874
##                                      P.Value    adj.P.Val
## Streptococcus_sp_HMSC070B10    1.983241e-09 1.608408e-06
## Desulfovibrio_legallii         7.235447e-05 2.262347e-02
## Acinetobacter_johnsonii        8.368732e-05 2.262347e-02
## Bifidobacterium_gallinarum     1.398604e-04 2.835670e-02
## Bifidobacterium_saeculare      2.625545e-04 4.258634e-02
## Providencia_rettgeri           3.720909e-04 4.964695e-02
## Brevibacterium_aurantiacum     4.603342e-04 4.964695e-02
## Brevibacterium_linens          5.213754e-04 4.964695e-02
## Escherichia_fergusonii         5.509526e-04 4.964695e-02
## Staphylococcus_warneri         7.065697e-04 5.730280e-02
## Fusobacterium_periodonticum    9.084294e-04 6.697602e-02
## Haemophilus_parainfluenzae     1.127537e-03 7.620274e-02
## Bifidobacterium_pullorum       1.592914e-03 9.937330e-02
## Leptotrichia_sp_oral_taxon_212 1.818263e-03 1.053294e-01
## Clostridium_sp_CAG_632         2.231543e-03 1.206521e-01
## Citrobacter_freundii           2.679407e-03 1.336037e-01
## Lactobacillus_johnsonii        2.800571e-03 1.336037e-01
## Kosakonia_pseudosacchari       4.019789e-03 1.811138e-01
## Citrobacter_youngae            5.174862e-03 2.061281e-01
## Streptococcus_massiliensis     5.254226e-03 2.061281e-01
```

```r
cat("\n=== Significant interaction species (FDR < 0.05) ===\n")
```

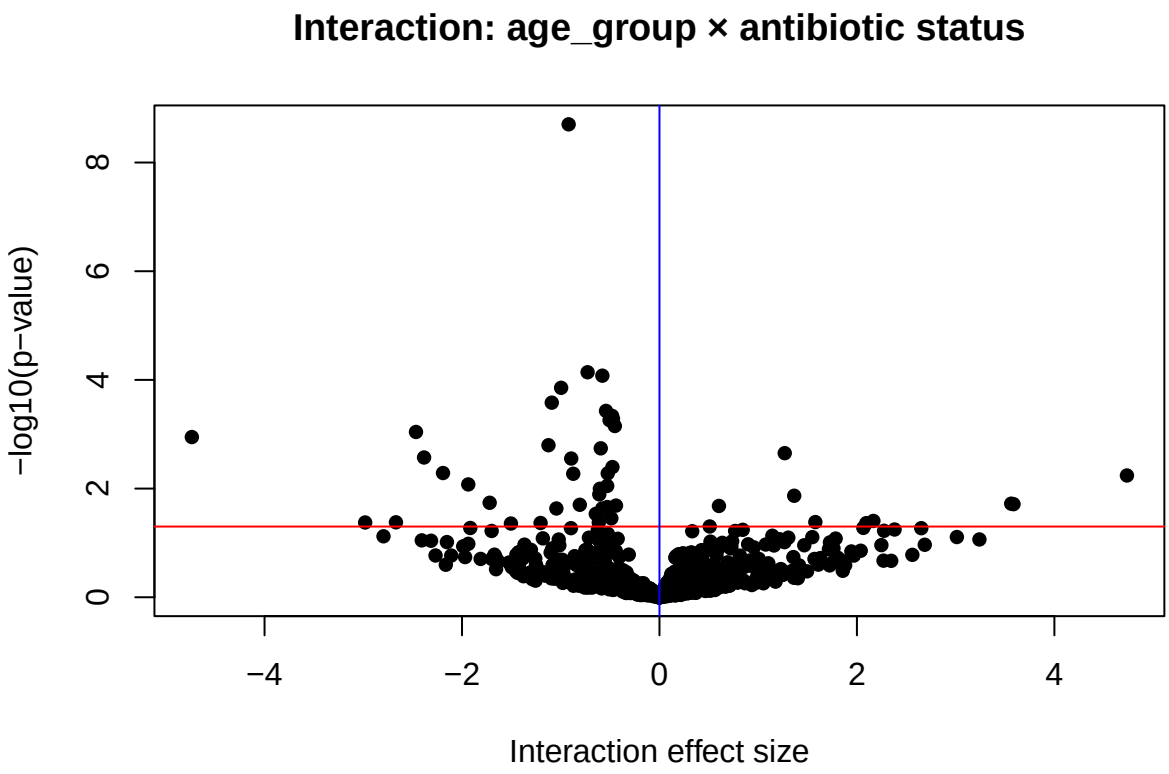
```
##
## === Significant interaction species (FDR < 0.05) ===
```

```r
print(sum(res_interaction$adj.P.Val < 0.05, na.rm = TRUE))
```

```
## [1] 9
```

```r
# ---- 17.8 Visualization ----
plot(
  res_interaction$logFC,
  -log10(res_interaction$P.Value),
  pch = 16,
  main = "Interaction: age_group × antibiotic status",
  xlab = "Interaction effect size",
  ylab = "-log10(p-value)"
)

abline(h = -log10(0.05), col = "red")
abline(v = 0, col = "blue")
```



Final Interpretation

Exploratory comparisons between antibiotic-controlled and uncontrolled studies revealed low concordance in estimated microbial effect sizes, suggesting instability in observed associations across study subsets. However, this approach is limited by reduced sample sizes and does not provide a formal statistical test of effect modification.

To address this limitation, an interaction-based modeling framework was applied. This analysis identified a subset of microbial species (n = 9, FDR < 0.05) for which the association between early and late onset colorectal cancer differed depending on antibiotic reporting status.

These findings provide statistical evidence that antibiotic-related study design differences may influence observed microbial signals. In particular, the presence of significant interaction terms indicates that the magnitude and, in some cases, direction of early versus late CRC associations vary across antibiotic reporting groups.

However, interpretation of these results requires caution. The number of early-onset CRC samples within antibiotic-controlled datasets was limited, leading to potential instability in interaction estimates. Additionally, imbalance across study cohorts may contribute to variability in estimated effects.

Taken together, these results suggest that microbial associations with early versus late colorectal cancer are sensitive to study composition and may be partially influenced by differences in antibiotic reporting. While this analysis provides evidence of effect modification, further validation in larger and more balanced cohorts with detailed patient-level antibiotic exposure data will be necessary to confirm these findings.

Overall, this study highlights the importance of accounting for antibiotic-related confounding and supports the use of interaction-based modeling approaches in microbiome research.